Deep Learning Course - First Project Report

Convolutional Neural Networks & Few-Shot Learning

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Dataset: CINIC-10

Why CINIC-10?

- Bridge between CIFAR-10 and ImageNet.
- 270,000 images total (90k train, 90k val, 90k test).
- 32×32 resolution.
- 10 balanced classes.

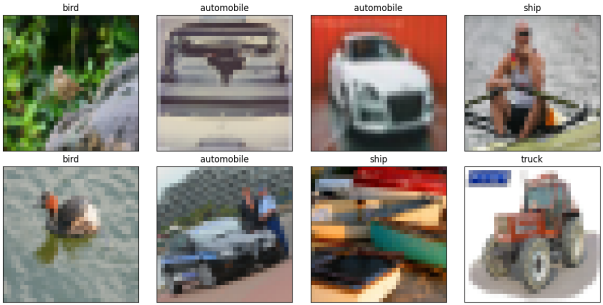


Figure: Sample raw images from CINIC-10

Data Augmentation Pipeline

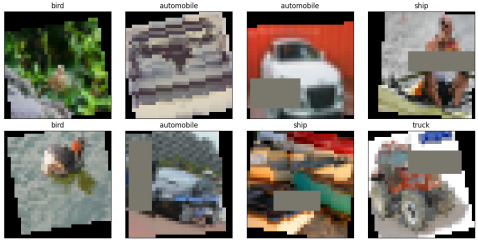


Figure: Native 32×32 (Custom, VGG-11)

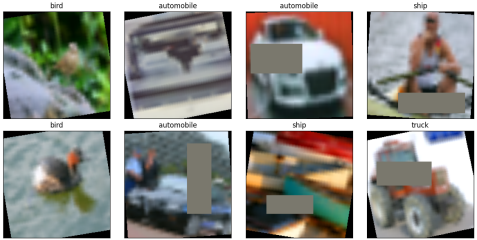


Figure: Resized 224×224 (ResNet, MobileNet)

Techniques used: Random Crop, Horizontal Flip, Rotation ($\pm 15°$), and Cutout.

Architectures: Scratch vs. Pre-trained

Custom CNN (Baseline)

Layer (Type)	Output Shape	Param #
Conv Blocks (x3)	$4 \times 4 \times 128$	93,568
Classifier	10	527,114
Total		**~620K**

ResNet-34 (Transfer Learning)

Layer (Type)	Status	Param #
Conv Backbone	Frozen	21,284,672
fc (New Head)	**Trainable**	5,130
Total		**~21.2M**

VGG-11 (From Scratch)

Layer (Type)	Output Shape	Param #
Conv Blocks (x8)	$1 \times 1 \times 512$	9,221,760
Dense Classifier	10	18,923,530
Total		**~28.1M**

MobileNetV2 (Transfer Learning)

Layer (Type)	Status	Param #
Inv. Residuals	Frozen	2,223,872
New Head	**Trainable**	12,810
Total		**~2.2M**

Global Leaderboard & Performance Gap

Rank	Model Configuration	Best Val Acc	Best Val Loss	Epoch	Overfit?
1	RESNET34_sgd_30_E_0.001_LR_0.0001_L2_Cutout_	0.911422	0.268619	25	No
2	RESNET34_sgd_30_E_0.001_LR_	0.906956	0.284914	14	No
3	RESNET34_adam_30_E_0.0001_LR_0.0001_L2_Cutout_	0.899633	0.312641	12	No
4	RESNET34_adam_30_E_0.0001_LR_	0.895167	0.321267	28	No
5	MOBILENET_sgd_30_E_0.001_LR_0.0001_L2_Cutout_	0.889667	0.327200	26	No
6	MOBILENET_adam_30_E_0.0001_LR_0.0001_L2_Cutout_	0.884944	0.344523	14	No
7	MOBILENET_sgd_30_E_0.001_LR_	0.883700	0.343569	25	No
8	MOBILENET_adam_30_E_0.0001_LR_	0.880378	0.357251	17	No
9	VGG11_sgd_30_E_0.01_LR_0.0_Drop_	0.736289	0.750715	28	No
10	VGG11_sgd_30_E_0.01_LR_0.5_Drop_	0.731867	0.768365	30	No
11	RESNET34_adam_30_E_0.001_LR_Frozen_0.0001_L2_	0.726233	0.786564	18	No
12	RESNET34_adam_30_E_0.001_LR_Frozen_	0.726167	0.786755	18	No
13	VGG11_sgd_30_E_0.01_LR_0.0_Drop_Cutout_	0.722822	0.781987	28	No
14	RESNET34_adam_30_E_0.001_LR_Frozen_Cutout_	0.722389	0.798848	24	No
15	VGG11_adam_30_E_0.0001_LR_0.0_Drop_	0.720711	0.831518	28	No
16	CNN_sgd_30_E_0.01_LR_0.0_Drop_	0.719667	0.796239	29	No
17	RESNET34_sgd_30_E_0.01_LR_Frozen_	0.716444	0.818318	12	No
18	CNN_adam_30_E_0.001_LR_0.0_Drop_0.0001_L2_	0.716089	0.799216	29	No
19	CNN_adam_30_E_0.001_LR_0.0_Drop_	0.715556	0.806796	29	No
20	CNN_sgd_30_E_0.01_LR_0.0_Drop_Cutout_	0.703922	0.833592	30	No
21	CNN_sgd_30_E_0.01_LR_0.5_Drop_	0.702889	0.835920	29	No
22	CNN_adam_30_E_0.001_LR_0.5_Drop_	0.693544	0.861760	29	No
23	CNN_adam_30_E_0.001_LR_0.3_Drop_0.0001_L2_Cutout_	0.691389	0.869183	30	No
24	VGG11_adam_30_E_0.001_LR_0.5_Drop_	0.676400	0.925652	27	No
25	CNN_adam_30_E_0.0001_LR_0.0_Drop_	0.676011	0.907821	29	No
26	VGG11_adam_30_E_0.001_LR_0.0_Drop_	0.649122	0.986749	27	No
27	VGG11_adam_30_E_0.001_LR_0.0_Drop_0.0001_L2_	0.648922	0.985195	28	No
28	MOBILENET_adam_30_E_0.001_LR_Frozen_Cutout_	0.634544	1.023366	26	No
29	VGG11_adam_30_E_0.001_LR_0.3_Drop_0.0001_L2_Cutout_	0.632733	1.030580	28	No
30	MOBILENET_sgd_30_E_0.01_LR_Frozen_	0.629367	1.042807	28	No
31	MOBILENET_adam_30_E_0.001_LR_Frozen_	0.628267	1.042022	13	No
32	MOBILENET_adam_30_E_0.001_LR_Frozen_0.0001_L2_	0.628000	1.043423	28	No

Global Leaderboard & Performance Gap

Model	Optimizer	Augmentation	Test Loss	Test Acc	Train Time
ResNet-34	SGD (0.001)	Crop+Flip+Rot+Cutout	0.3223	**0.9103**	292.95 min
MobileNetV2	SGD (0.001)	Crop+Flip+Rot+Cutout	0.3440	**0.8893**	243.23 min
VGG-11	SGD (0.01)	Crop+Flip+Rot	0.7936	**0.7257**	19.71 min
Custom CNN	SGD (0.01)	Crop+Flip+Rot	0.8331	**0.7067**	70.79 min

Training Dynamics: Transfer Learning

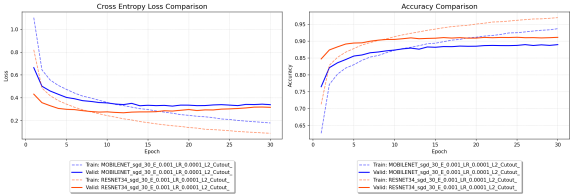


Figure: MobileNet vs ResNet-34

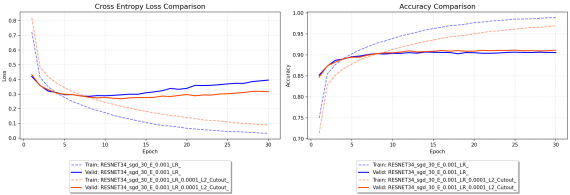


Figure: Impact of Cutout on ResNet-34

Training Dynamics: Models Trained from Scratch

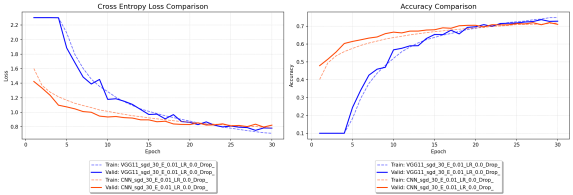


Figure: VGG-11 vs Custom CNN

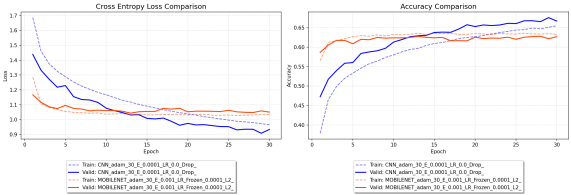


Figure: Scratch vs Transfer Learning

Ablation Studies: VGG-11 Hyperparameters

Rank	Model Configuration	Best Val Acc	Best Val Loss	Epoch	Overfit?
1	VGG11_sgd_30_E_0.01_LR_0.0_Drop_128_BS_	0.736289	0.750715	28	No
2	VGG11_adam_30_E_0.0001_LR_0.0_Drop_128_BS_	0.720711	0.831518	28	No
3	VGG11_sgd_30_E_0.01_LR_0.5_Drop_0.0001_L2_128_BS_Cutout_	0.718478	0.797689	28	No
4	VGG11_adam_30_E_0.001_LR_0.0_Drop_128_BS_NoRot_	0.713756	0.834615	30	No
5	VGG11_adam_30_E_0.001_LR_0.5_Drop_128_BS_	0.676400	0.925652	27	No
6	VGG11_adam_30_E_0.001_LR_0.0_Drop_128_BS_Cutout_	0.675567	0.919039	30	No
7	VGG11_adam_30_E_0.001_LR_0.0_Drop_256_BS_	0.675411	0.910392	29	No
8	VGG11_adam_30_E_0.001_LR_0.7_Drop_128_BS_	0.660367	0.975262	30	No
9	VGG11_adam_30_E_0.001_LR_0.0_Drop_128_BS_NoCrop_NoRot_	0.651078	1.031305	16	Yes
10	VGG11_adam_30_E_0.001_LR_0.3_Drop_128_BS_	0.650856	0.984784	28	No
11	VGG11_adam_30_E_0.001_LR_0.0_Drop_128_BS_	0.649122	0.986749	27	No
12	VGG11_adam_30_E_0.001_LR_0.0_Drop_0.0001_L2_128_BS_	0.648922	0.985195	28	No
13	VGG11_adam_30_E_0.001_LR_0.0_Drop_64_BS_	0.648000	0.995910	30	No
14	VGG11_adam_30_E_0.001_LR_0.3_Drop_0.0001_L2_128_BS_Cutout_	0.632733	1.030580	28	No
15	VGG11_sgd_30_E_0.1_LR_0.0_Drop_128_BS_	0.629011	1.101426	16	No
16	VGG11_adam_30_E_0.001_LR_0.0_Drop_128_BS_NoCrop_NoFlip_NoRot_	0.607300	1.128019	14	Yes
17	VGG11_adam_30_E_1e-05_LR_0.0_Drop_128_BS_	0.527444	1.275207	28	No

Figure: Ranking of VGG-11 configurations (LR, Batch Size, Optimizers, Regularization)

Hyperparameter Insights: LR & Batch Size

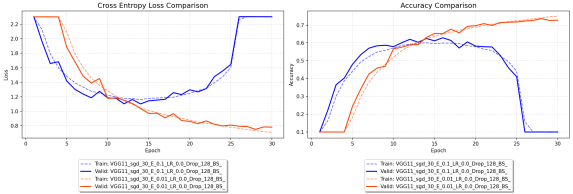


Figure: SGD: 0.1 vs 0.01 (Instability of high LR)

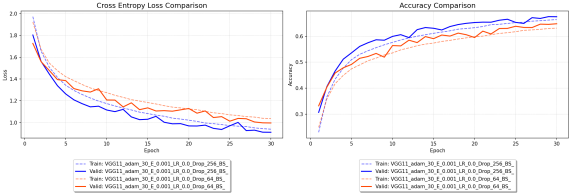


Figure: Batch Size: 256 vs 64 (Faster convergence)

Hyperparameter Insights: Regularization

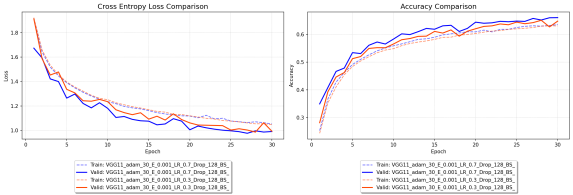


Figure: Dropout: 0.7 vs 0.3

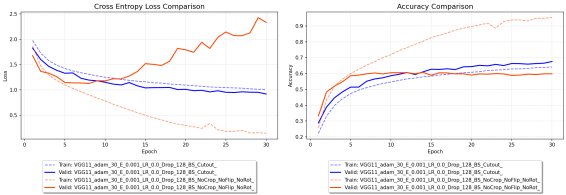
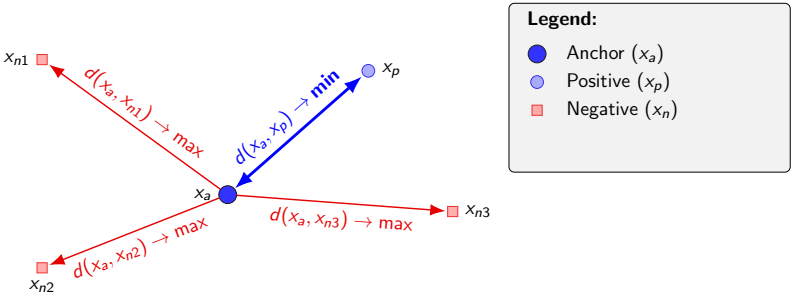


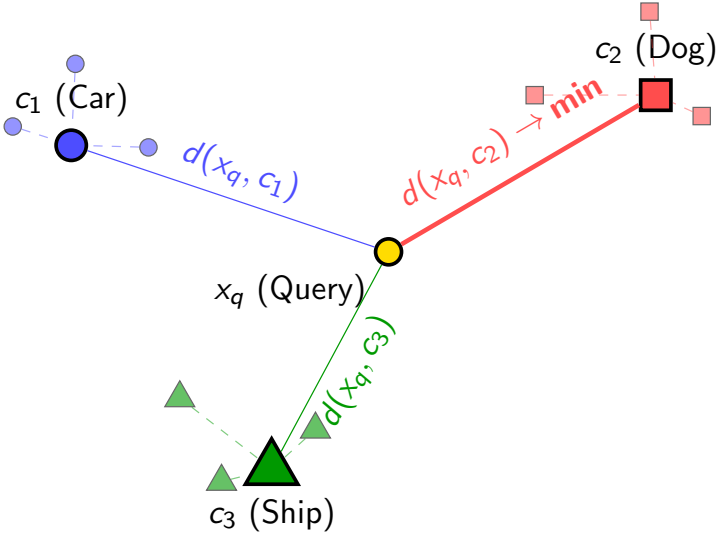
Figure: Cutout vs No Augmentation

Few-Shot: Contrastive Learning (Triplet Loss)



$$L(A, P, N) = \max(0, \|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha)$$

Few-Shot: Prototypical Networks



Few-Shot Learning Results

Rank	Method	Backbone	Shots	Tuning	Hyperparameters	Aug	Val Acc	Test Acc
1	ProtoNet	ResNet-34	50	Full FT	emb: 128, drop: 0.3	Standard	0.6784	0.6759
2	ProtoNet	ResNet-34	50	Frozen	emb: 128, drop: 0.3	Standard	0.6380	0.6337
3	ProtoNet	ResNet-34	50	Frozen	emb: 128, drop: 0.3	Cutout	0.5966	0.5919
4	ProtoNet	ResNet-34	10	Full FT	emb: 128, drop: 0.3	Standard	0.5287	0.5282
5	ProtoNet	ResNet-34	10	Full FT	emb: 256, drop: 0.3	Standard	0.5174	0.5123
6	ProtoNet	ResNet-34	10	Frozen	emb: 128, drop: 0.3	Standard	0.5086	0.5059
7	ProtoNet	ResNet-34	10	Full FT	emb: 64, drop: 0.3	Standard	0.4998	0.4941
8	ProtoNet	ResNet-34	10	Full FT	emb: 128, drop: 0.5	Cutout	0.4534	0.4358
9	ProtoNet	ResNet-34	5	Full FT	emb: 128, drop: 0.3	Standard	0.4396	0.4398
10	Contrastive	ResNet-34	50	Frozen	Margin: 1.0	Standard	0.4372	0.4325
11	Contrastive	ResNet-34	10	Frozen	Margin: 1.0	None	0.4091	0.3617
12	Contrastive	MobileNet	10	Frozen	Margin: 1.0	None	0.3976	0.3165
13	Contrastive	ResNet-34	10	Frozen	Margin: 1.0	Standard	0.3799	0.3583
14	Contrastive	ResNet-34	10	Frozen	Margin: 2.0	Standard	0.3799	0.3488
15	Contrastive	ResNet-34	10	Frozen	Margin: 1.0, WD: 1e-4	Standard	0.3798	0.3584
16	Contrastive	MobileNet	50	Frozen	Margin: 1.0	Standard	0.3607	0.3518
17	Contrastive	ResNet-34	10	Frozen	Margin: 1.0	Cutout	0.3571	0.3571
18	Contrastive	MobileNet	10	Frozen	Margin: 1.0, WD: 1e-4	Standard	0.3546	0.3334
19	Contrastive	MobileNet	10	Frozen	Margin: 2.0	Standard	0.3544	0.3258
20	Contrastive	MobileNet	10	Frozen	Margin: 1.0	Standard	0.3432	0.3317
21	Contrastive	MobileNet	10	Frozen	Margin: 1.0	Cutout	0.3348	0.3201

Few-Shot Learning Results

Key Findings:

- Prototypical Networks significantly outperformed Contrastive Learning in our setup.
- 50-shot learning achieved up to 67.59% test accuracy.
- Fine-tuning the ResNet-34 backbone (Full FT) yielded better results than keeping it frozen.

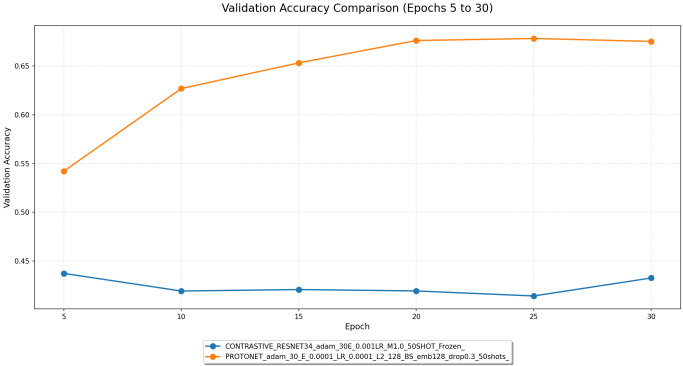


Figure: ProtoNet vs Contrastive

Summary & Conclusions

- **Transfer Learning is King:** Pre-trained models (ResNet-34) massively outperformed scratch models, hitting 91% accuracy, albeit at a higher computational cost.
- **Data Augmentation is Non-Negotiable:** Spatial transforms and Cutout were critical in preventing early overfitting across all architectures.
- **Optimizer Tuning:** Adam provided fast convergence, while SGD was highly sensitive to the initial learning rate.
- **Metric Learning:** Prototypical Networks proved to be an elegant and effective solution for Few-Shot classification compared to standard Triplet Loss setups.

Thank You!